

X-Ray-Based Lightweight High-Performance Non-Destructive Defect Detection Method for High-Voltage Cables

Junyu Chen¹, Beiping Hou^{1*†}, Jianwei Dong^{1†}, Wen Zhu^{1†},
Fangkun Shao^{1†}, Jing Jie^{1†}

^{1*}College of Automation and Electrical Engineering, Zhejiang University of Science and Technology, Hangzhou, 310023, China.

*Corresponding author(s). E-mail(s): bphou@zust.edu.cn;
Contributing authors: chenjunyu020503@163.com;
dongjw_edu@163.com; zhuwen@zust.edu.cn; 547124542@qq.com;
jingjie@zust.edu.cn;

[†]These authors contributed equally to this work.

Abstract

High-voltage cable defect detection is an important part of urban power system maintenance. Due to variations in cable thickness, large weight, and challenging operability, existing signal-processing and artificial-intelligence-based methods may face trade-offs among detection accuracy, cost, efficiency, and applicability. Moreover, some existing methods are invasive in nature. Therefore, this paper proposes an improved multiscale attention-based YOLOv8 model, designs a portable X-ray non-destructive testing (NDT) system, and creates a larger dataset based on the CycleGAN-based (Cycle Consistent Adversarial Networks-based) Pix2Pix (Pixel-to-Pixel Translation) model. First, the improved YOLOv8 model introduces EfficientNet to replace CSPNet (Cross Stage Partial Network) as the backbone network and incorporates a Squeeze-and-Excitation (SE) module after each convolutional block. Second, a bottom-up feature pyramid PAN (Path Aggregation Network) is integrated with the FPN (Feature Pyramid Network) to construct a hybrid multi-scale feature fusion structure, enabling multi-level and multi-directional information transfer. Moreover, a task-specific box refinement strategy combining K-means initialization with IoU-guided matching is employed to improve localization accuracy. The experimental results show that, compared with the other detection models, EMA-YOLOv8 improves the average precision and mAP50 by 4.68 and 3.9 percentage points, respectively, while reducing the

average number of parameters and GFLOPs by 35.37% and 27.76%, respectively. Therefore, the proposed model has great potential.

Keywords: High-voltage cable non-destructive defect detection, X-ray image processing, Non-destructive testing equipment, YOLOv8

1 Introduction

With the acceleration of economic development and urban modernization, there is an increasing demand for the safety, reliability and efficiency of power transmission[1][2]. As the core component of modern urban power system, high-voltage cable undertakes the task of transmitting huge current. However, high-voltage cables with long-term high load operation face potential risks such as material aging, insulator damage, and conductor breakage[3], as shown in Fig. 1. These problems will not only lead to the failure of the power system, but also may cause serious safety accidents, such as fire, casualties and so on. Therefore, the implementation of daily maintenance and regular inspection of high-voltage cables is the core guarantee to ensure the safe and stable operation of the power system.



Fig. 1 High-voltage cable defect images.

At present, there are two methods for daily inspection of cables: signal processing[4–6] and artificial intelligence[7–10]. However, both methods have limitations. For signal processing methods, [11], [12] and [13] used the transmission characteristics of ultrasonic signals in cables to detect defects, but this method is technically complex and has high requirements for contact media, which is not conducive to industrial popularity. [14], [15] and [16] respectively used structured light and partial discharge methods to detect cable defects, but these two methods are very sensitive to changes in environmental factors, and it is difficult to accurately detect in strong light or high noise environment. [17] established a microwave reflection theoretical model

to detect cable defects, but due to the heavy insulation of high-voltage cables, the microwave reflection ability is limited, and it is impossible to measure them with high precision. At the same time, there are some other signal processing methods to detect cable defects, such as Kalman filtering[18], time-frequency domain filtering[19], singular value decomposition[20] and autoencoder[21]. These methods need a large number of complex mathematical calculations, which do not meet the requirements of high efficiency in industrial detection. Therefore, it is urgent to develop a technical solution suitable for multi-type high-voltage cables to realize efficient and low-cost defect detection in complex environments.

In recent years, artificial intelligence is more and more widely used in cable defect detection, which can effectively reduce the shortcomings of traditional signal processing methods[22][23]. [24] proposed a defect detection method based on random forest combined with fluorescence penetration. Although this method shows certain capability on small training sets, the model accuracy is low due to lack of data. [25] proposed a defect detection method combining image enhancement technology and principal component analysis. Although the method designs a closed box to control the brightness of the detection area, for high-voltage cables located in the air, the model generalization ability is still low due to bad weather. In addition, many image recognition neural network models such as CNN[26], MemSeg[27], YOLO[28], etc., were also widely used in defect detection. However, despite the significant improvements in detection accuracy achieved by the aforementioned methods, several critical limitations remain. These include the lack of versatility in network models, high hardware requirements, substantial computational resource consumption, and the destructive nature of most techniques on the cables themselves. These constraints hinder their ability to meet the growing demands of rapidly advancing industrial inspection technologies.

Aiming at the limitations of traditional signal processing methods in environmental adaptability and detection accuracy, as well as the technical bottlenecks of artificial intelligence that have high accuracy but lack versatility, ease of use and non-destructive testing, some scholars combined X-ray with artificial intelligence such as classification neural networks and Skip-GANomaly to detect cable defects[29][30]. Anomaly detection or novelty detection like Skip-GANomaly in [30] are suitable when data are limited or when unknown defects need to be identified, but they may not directly provide explicit defect categories and bounding boxes. Since different defect types are predefined and both classification and localization are required, supervised object detection is adopted. Furthermore, X-ray has many advantages, including high penetration, non-destructive testing and high resolution[31], it has been widely used in PCB board defect identification[32], logistics cargo transportation[33], chemical substance detection[34] and other fields. In addition, as a lightweight network, YOLOv8 model has high precision and low computational cost[35], and some scholars have widely applied it in industrial testing. For example, [36] proposed an improved YOLOv8 model to detect bearing defects and [37] proposed an improved YOLOv8 model to detect crane operation tools. Therefore, based on the advantages of X-ray and YOLOv8 model, this paper proposes a lightweight high-performance non-destructive defect detection method in high-voltage cables named EMA-YOLOv8. The main contributions of this paper can be summarized as follows:

1. The CSPNet backbone of the original YOLOv8 network is replaced with EfficientNet, and an SE module is incorporated after each Conv-BN-ReLU (Convolution-Batch Normalization-Rectified Linear Unit) block to adaptively adjust channel responses. In addition, a hybrid pyramid feature fusion module integrating FPN and PAN is introduced to enhance multi-scale feature representation. On this basis, a task-specific box refinement strategy combining K-means initialization with IoU-guided matching is employed to improve localization accuracy. These modifications collectively enhance the model's ability to extract and fuse defect features at different scales, thereby improving the detection performance for high-voltage cable X-ray defects.
2. To address the issue of insufficient model training accuracy due to limited datasets, a CycleGAN-based Pix2Pix model is proposed, which effectively augments the dataset and enhances training accuracy.
3. A portable suite of X-ray-based NDT equipment has been developed, enabling rapid deployment to required testing locations.

2 X-Ray Data Acquisition and Dataset Preparation

This paper uses a self-designed X-ray equipment to construct a comprehensive dataset containing 1000 X-ray images of high-voltage cable defects. Based on the distance of the damaged part from the battery core, the damage was divided into four types: insulation damage, foreign body damage, insulation-conductor damage and conductor-foreign body damage. To improve the reliability and accuracy of the model, 40 low-quality images are manually excluded. Consequently, a total of 960 high-quality X-ray defect images of high-voltage cables are retained, including 600 single defect images and 360 multi-defect images, of which 310 are insulator damage, 195 are foreign body damage, 145 are insulator-conductor damage, and 150 are conductor-foreign body damage. All data labels were created by the authors using Labellmg, ensuring consistent labeling quality and overall consistency.

2.1 Equipment Introduction

To better acquire X-ray data from high-voltage cables, this paper designs a acquisition equipment. Table 1 is the specific parameters of the high-voltage cable NDT equipment respectively. The entire system is divided into two parts: X-ray imaging system and X-ray image processing system, as shown in Fig. 2. The X-ray imaging system consists of a portable X-ray emitter and an X-ray receiving board. Due to the density and thickness differences in different areas inside the cable, when the x-rays reach the receiving board, the Flat Panel Detector in the receiving board converts the intensity differences of these X-rays into electrical signals, and further processes them to generate a grayscale image. These images are transmitted to the image processing system through the serial port, then the CycleGAN-based Pix2Pix model and EMA-YOLOv8 algorithms are used for further enhancement and defect detection.

The compact and lightweight design of the entire device makes it easy to deploy and operate in complex field environments, especially for mission scenarios that require quick installation and movement. When detecting cable defects, first connect the X-ray

Table 1 Parameter of the high-voltage cable NDT equipment.

Parameters	Values
Apparent dimension of X-ray emitter	$8 \times 8 \times 30\text{cm}$
Main engine weight of X-ray emitter	3KG
Type of focus of X-ray emitter	microfocus
Viewing angle range of X-ray emitter	$\phi 50\text{mm}$
Tube voltage of X-ray emitter	[65KV, 100KV]
Target angle of X-ray emitter	12
Gray level of X-ray emitter	Level 8
Light leakage of X-ray emitter	$<1\text{mR/h}$
Apparent dimension of X-ray receiving board	$30 \times 10 \times 30\text{cm}$
Main engine weight of X-ray receiving board	8KG
Drawing speed of X-ray receiving board	Real-time mapping
Resolution of X-ray receiving board	801p/cm^2
Contrast of X-ray receiving board	7%

emitter to the power supply and connect the X-ray receiving board to the computer, turn on the power switch, manually adjust the voltage value according to the thickness of the high-voltage cable, then place the cable under test between the transmitter and the receiving board, press the start button on the transmitter, and the X-ray image of the cable under test can be displayed in real time on the terminal connected to the transmitting interface of the receiving board. Finally, the image processing system detects the defects of the high-voltage cable.

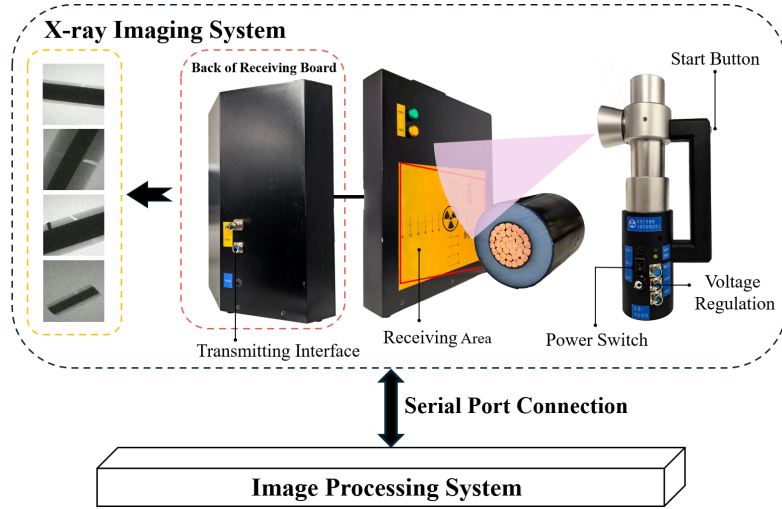


Fig. 2 High-voltage cable NDT equipment.

2.2 Image Denoising and Enhancement

The X-ray images acquired using the equipment described in Section 2.1 may contain noise and low local contrast, which can obscure defect boundaries and fine structural information. Since the amount of real X-ray data is limited, variations in image quality may affect feature extraction and detection stability. Therefore, denoising and contrast enhancement are applied to improve defect visibility and provide clearer inputs for subsequent feature extraction.

FFDNet is a computationally efficient denoising network designed for high-resolution image processing. It partitions input images into smaller patches, which are processed independently through an optimized convolutional architecture to enhance computational speed and performance.

After acquiring the denoised X-ray images, the problems of low contrast and unclear image details still exist. To solve these problems, adaptive histogram equalization techniques are used, which can be applied to different regions of the image, significantly optimizing the contrast and brightness. Fig. 3 shows the result of applying denoising and enhancement techniques to the original X-ray images.

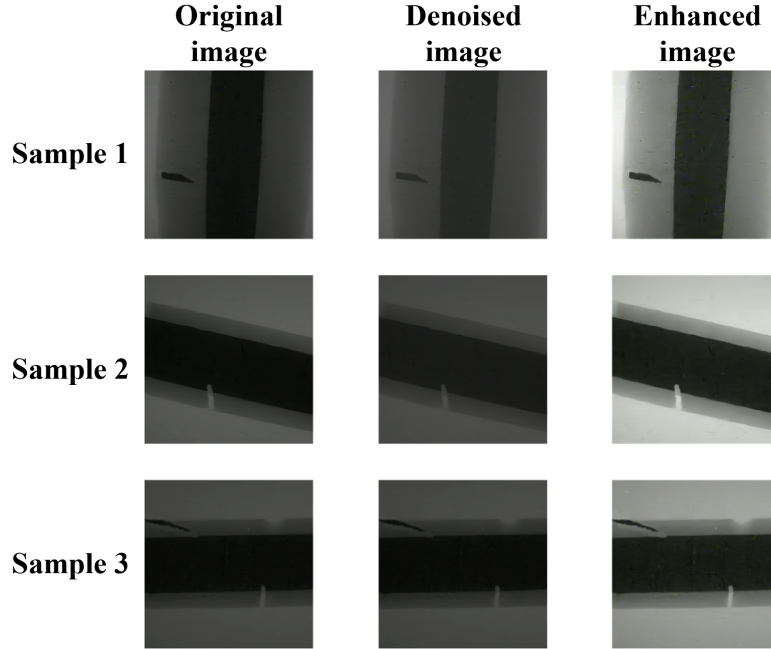


Fig. 3 Comparison of high-voltage cable X-ray images after denoising and enhancement.

2.3 Dataset Expansion

Although these images encompass a broad spectrum of defect types, the overall data volume remains insufficient to train a deep learning model with robust generalization

capabilities. Therefore, this paper proposes a CycleGAN-based Pix2Pix model for dataset expansion.

The CycleGAN-based Pix2Pix model is a conditional generative adversarial network (cGAN) variant. Its architecture integrates conditional inputs into both the generator and discriminator, enabling the generator to produce high-fidelity target images based on input images. The generator employs a modified U-Net structure: the encoder extracts features through convolutional layers and max-pooling layers for downsampling, while the decoder reconstructs resolution via transposed convolutions and bilinear interpolation. To enhance global coherence, a self-attention mechanism is incorporated into the upsampling phase, capturing long-range dependencies to refine critical region modeling and improve complex detail generation. The discriminator adopts a PatchGAN framework, partitioning input images into local patches for independent evaluation via convolutional networks. Each patch’s probability of authenticity is averaged to compute the overall image authenticity, balancing local texture fidelity and computational efficiency. Through adversarial training, the generator progressively refines output realism, while the discriminator sharpens its discriminative capability, achieving synergistic optimization of the model’s performance. Fig. 4 shows the schematic diagram of CycleGAN-based Pix2Pix model.

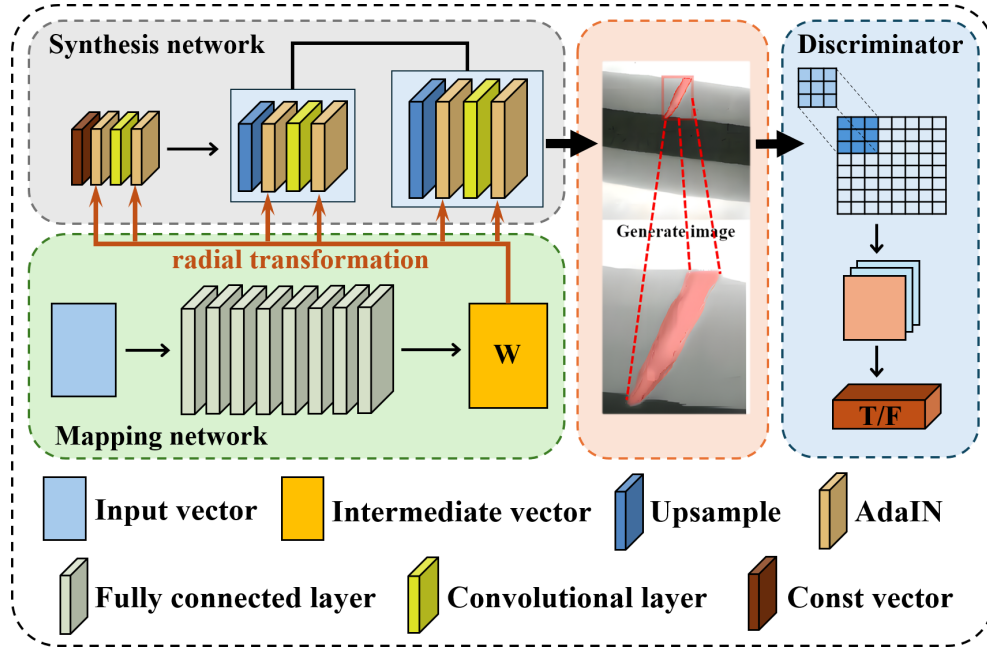


Fig. 4 Schematic diagram of CycleGAN-based Pix2Pix model.

The CycleGAN-based Pix2Pix model generated 1022 new images. The proportion of various defects remained basically consistent with the original images. To further evaluate the distribution similarity, diversity, and feature distribution of the generated

data, FID, LPIPS, and t-SNE analyses are conducted. The FID of the generated images is 11.8. The mean LPIPS between the real images and their corresponding synthetic images is 0.739, with a standard deviation of 0.283, indicating noticeable perceptual variations rather than simple duplication of the original samples. As shown in Fig. 5, the synthetic and real samples overlap in several major feature regions, while the synthetic samples exhibit a broader distribution. These results indicate that the generated images preserve important characteristics of the real X-ray data while introducing additional diversity, although a certain degree of distribution difference remains. Part of the generated images are shown in Fig. 6.

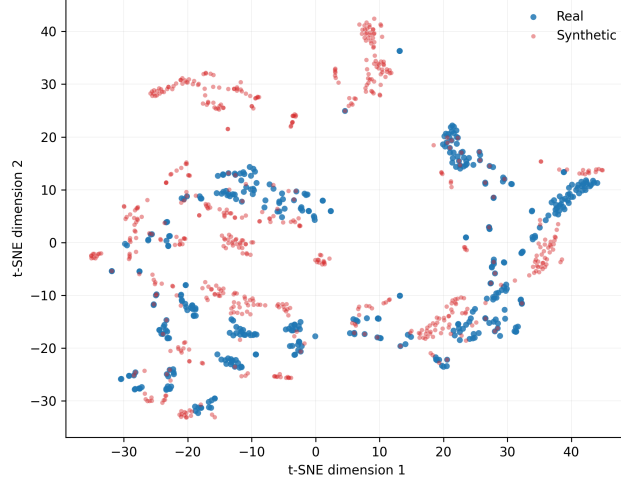


Fig. 5 t-SNE visualization of the feature distributions of real and synthetic X-ray images.

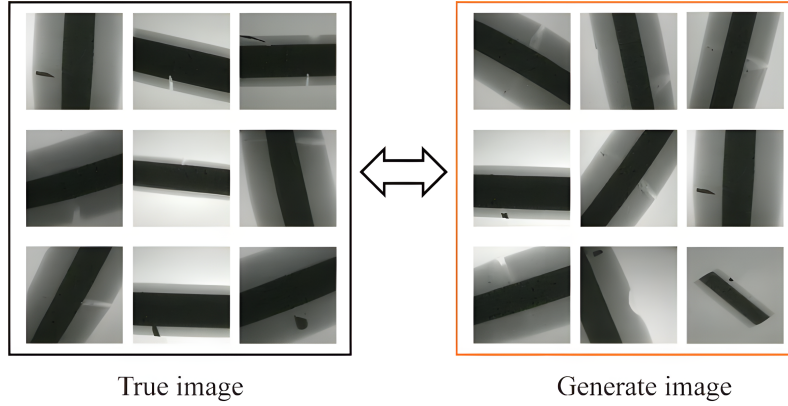


Fig. 6 True dataset and generated dataset.

3 High-voltage Cable Defect Detection

This section will concentrate on the method for detecting defects in high-voltage cables. The original YOLOv8 model may exhibit limitations in feature extraction and multi-scale representation when detecting cable defects with different sizes and shapes. Therefore, this section introduces the modifications adopted in EMA-YOLOv8 to improve feature representation and localization performance.

3.1 YOLOv8 Model

YOLOv8 is a detection framework combining efficiency and precision capabilities. The system architecture comprises four principal components: an input preprocessing module, a CSPNet-based backbone network, a multi-scale feature fusion neck, and a decoupled prediction head.

The input pipeline employs adaptive image standardization via bicubic interpolation for resolution normalization and scales pixel values to $[0,1]$ or $[-1,1]$ ranges to ensure numerical stability during network computation, after which the preprocessed tensor is processed by a CSPNet-53 backbone utilizing cross-stage partial connections to extract hierarchical feature representations while maintaining computational efficiency. The feature enhancement network integrates PAN with modified spatial pyramid pooling (SPPF) modules to enable bidirectional cross-scale feature fusion that preserves both semantic richness and spatial granularity, and the output layer implements a decoupled head architecture with task-specific convolutional branches for coordinate regression, class probability estimation, and confidence prediction, followed by non-maximum suppression with optimized IoU thresholds for the final detection output.

However, while YOLOv8 maintains high detection efficiency and real-time performance, its larger model versions may limit deployment flexibility in resource-constrained environments due to the high computational load requirements.

3.2 EfficientNet Model

Traditional network scaling methods typically focus on expanding only one dimension—either depth, width, or resolution. On the contrary, EfficientNet achieves a good balance between computational complexity and accuracy by uniformly scaling all three dimensions[38], as shown in Fig. 7. This composite scaling strategy can be mathematically expressed as follows:

$$d = \alpha^\varphi, w = \beta^\varphi, r = \gamma^\varphi \quad (1)$$

where d , w and r represent the depth, width and resolution of the network respectively; α , β and γ are constants used to determine the scaling ratio; and φ is a free variable used to control the allocation of computing resources for the entire network.

3.3 IoU-Guided Box Matching and Refinement

To improve the localization accuracy of high-voltage cable defects, this paper employs a task-specific box refinement strategy that combines K-means-based initialization with

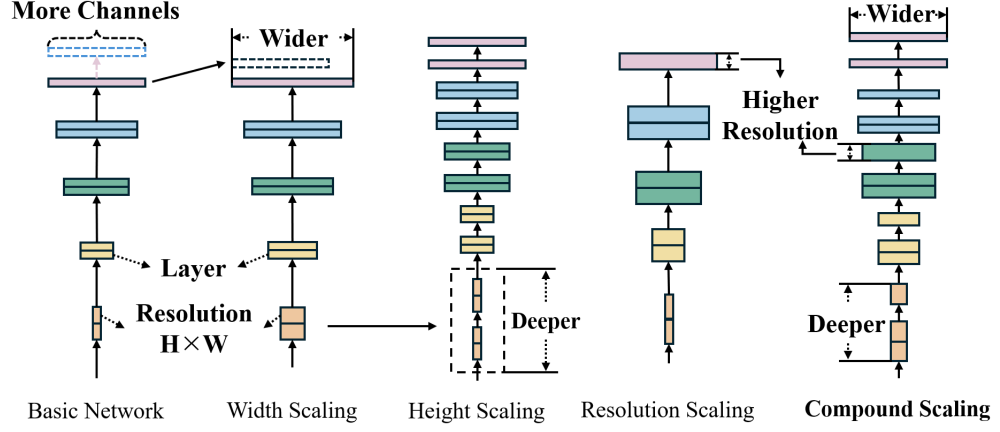


Fig. 7 EfficientNet compound scaling.

IoU-guided matching. Specifically, K-means clustering is first used to obtain initial box priors according to the size and aspect-ratio distribution of the ground-truth defect boxes. The IoU between each candidate box and the corresponding ground-truth box is then calculated to evaluate their matching degree. Candidate boxes with higher IoU values are selected as positive matches and are further refined according to the matching results. This procedure enables the candidate boxes to better adapt to the spatial distribution of cable defects. The overall refinement process is illustrated in Fig. 8.

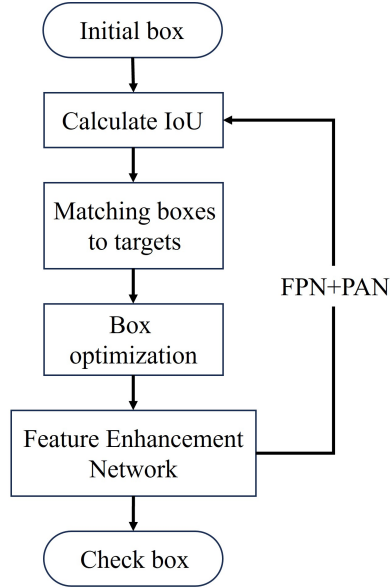


Fig. 8 Flowchart of the IoU-guided box refinement strategy.

The matching criteria can be expressed as:

$$IoU(b_i, a_j) = \frac{Area(b_i \cap a_j)}{Area(b_i \cup a_j)} \quad (2)$$

When $IoU(b_i, a_j)$ exceeds the set threshold of 0.8, the candidate box a_j and the ground-truth box b_i are considered a matched pair.

To accommodate defects with different sizes, K-means initialization is used to generate box priors covering a range of target scales. These initial boxes are subsequently matched and refined according to IoU, allowing their positions and scales to better adapt to the spatial characteristics of different cable defects. This process improves the localization capability of the model for defects at different scales.

3.4 Optimized Multi-Scale Feature Fusion

Based on the FPN layer of YOLOv8, a bottom-up feature pyramid PAN is added, and a hybrid pyramid structure is designed. Different levels of feature maps are systematically grouped and integrated according to a predefined strategy. This facilitates multi-level and multi-direction information flow. The hybrid pyramid structure effectively coordinates the features of different scales and improves the detection accuracy of the model. A graphical representation of this structure is shown in Fig. 9.

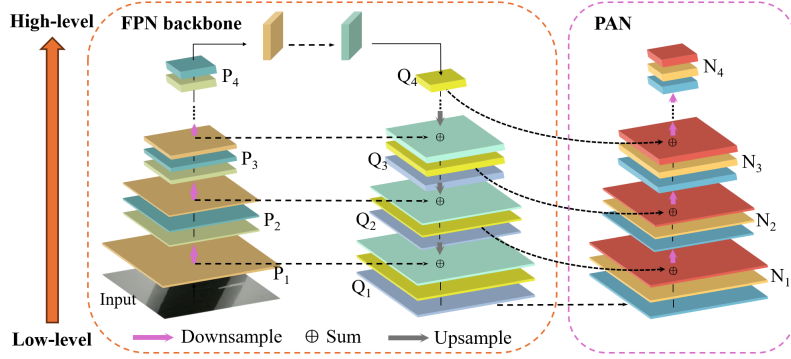


Fig. 9 Hybrid pyramid structure.

Assuming that feature maps P_1 , P_2 , P_3 , and P_4 correspond to features of different scales, after fusion using FPN:

$$Q_i = Conv(Upsample(P_{i+1}) + P_i) \quad (3)$$

where $Upsample$ is the up sampling operation, $Conv$ is the convolution operation, and Q_i is the fused feature map.

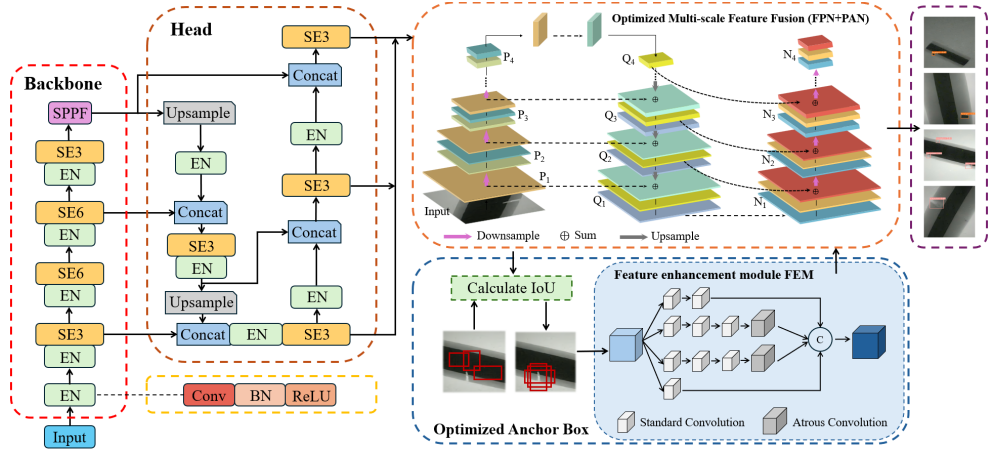


Fig. 10 EMA-YOLOv8 network structure.

By introducing bidirectional fusion, it can be expressed as:

$$N_i = \text{Conv}(\text{Upsample}(P_{i+1}) + P_i + \text{Downsample}(P_{i-1})) \quad (4)$$

where *Downsample* is a down sampling operation.

The deep feature maps carry stronger semantic features and weaker localization information, while the shallow feature maps carry stronger location information and weaker semantic features. FPN transfers the semantic features from deep layers to shallow layers to enhance the semantic expression at multiple scales, while PAN transfers the localization information from shallow layers to deep layers to enhance the localization ability at multiple scales.

3.5 A High-Voltage Cable Defect Detection Method Based on EMA-YOLOv8

This paper proposes a novel EMA-YOLOv8 model. The whole network has four modules, which are the backbone network, the head network, the optimized multi-scale feature fusion and the box refinement strategy. As shown in Fig. 10, EMA-YOLOv8 leverages EfficientNet as its fundamental backbone for feature extraction, seamlessly integrating the SE module following each Conv-BN-ReLU convolutional operation. This setup enables adaptive adjustment of channel weights through the squeeze-and-excitation operations. Following the network head comes the optimized the multi-scale fusion module, a hybrid pyramid structure is constructed on the traditional FPN and PAN. Through up-sampling, down-sampling and feature fusion, the feature map of the same level of the previous pyramid is fused with the feature map of the previous level, so as to realize multi-level and multi-directional information transfer.

Furthermore, to improve the performance of high-voltage cable defect detection, this paper introduces a task-specific box refinement strategy. Specifically, the size and aspect ratio of the initial box priors are determined using K-means clustering, and IoU-guided matching is then employed to evaluate the matching degree between the candidate boxes and the ground-truth boxes. Candidate boxes that do not satisfy the predefined matching criterion are further adjusted in position and scale, and the IoU is recalculated until the matching threshold of 0.8 is satisfied. In addition, the feature maps are passed to the multi-scale feature fusion module through the FEM (Feature Enhancement Module). The FEM module enriches the feature representation through a multi-branch dilated convolution structure, expands the local receptive field of the network, and enhances the semantic representation of small targets.

3.6 Optimized Loss Function

The traditional YOLO series models usually use a three-part loss function during training, namely bounding box regression loss, classification loss, and confidence loss. In order to better meet the needs of high-voltage cable defect detection, this paper optimizes the loss function of YOLOv8, which is specifically reflected in the following aspects:

1. In terms of bounding box regression loss, a generalized CIoU (Complete IoU) loss is introduced, which takes into account factors such as IoU, center point distance, and aspect ratio. The formula of CIoU loss is as follows:

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^g)}{c^2} + \alpha \cdot \nu \quad (5)$$

where $\rho(b, b^g)$ represents the Euclidean distance between the center point of the predicted box and the true box, c represents the diagonal length of the minimum closure area containing the predicted box and the true box, and α and ν are balancing factors designed to adjust the aspect ratio of the predicted box.

2. In terms of classification loss, this paper uses Focal Loss to replace the traditional cross entropy loss. Focal Loss can solve the problem of class imbalance by adjusting the weight of samples and focusing on samples that are difficult to classify. The formula is as follows:

$$L_{cls} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (6)$$

where p_t represents the probability predicted by the model, and α_t is a hyperparameter used to control the weights of easy-to-classify samples and difficult-to-classify samples.

3. In terms of confidence loss, this paper introduces an adaptive weight adjustment mechanism to further improve the detection capability of small target defects. In this way, the model can more accurately distinguish between real defects and background noise, improving the robustness of detection. The formula is as follows:

$$L_{conf} = -\sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (7)$$

Finally, the overall loss function of YOLOv8 is the weighted sum of the above three losses:

$$L = \lambda_{CIoU} L_{CIoU} + \lambda_{cls} L_{cls} + \lambda_{conf} L_{conf} \quad (8)$$

where λ_{CIoU} , λ_{cls} and λ_{conf} are the weight coefficients of the three losses.

4 Experimental Results and Analysis

4.1 Experimental Settings

This experiment trains and deploys the model on a device with a GPU model of NVIDIA GeForce RTX 3090 Ti (24GB video memory). The operating system is Ubuntu 18.04, the CUDA version is 11.4, and the Python version is 3.9.16.

During the model training procedure, the dataset was meticulously partitioned into a training set, a validation set, and a test set, with a ratio of 7:2:1. To ensure efficient training, the Adam optimizer was adopted, with the momentum parameter configured to 0.9. Furthermore, the training process was delineated into two distinct stages: the initial learning rate stage and the learning rate decay stage. In the initial learning rate stage, the learning rate was set at 0.001. Conversely, in the learning rate decay stage, the learning rate was progressively decreased, ensuring that it never dipped below 0.01 times the initial learning rate. The entire training process was designated to run for 600 iterations and set batch size to 16.

4.2 Expanded Dataset Effect Validation

The CycleGAN-based Pix2Pix model generates 1,022 additional X-ray images of cable defects, expanding the dataset from 960 real images to a total of 1,982 images. The synthetic images are used only to augment the training data, while the validation and test sets contain only real X-ray images. To evaluate the effect of dataset expansion, the EMA-YOLOv8 model trained with the expanded training set is compared with the same model trained using only the original real images under the same experimental settings described in Section 4.1. Four defect categories are considered for evaluation, with 20 real images selected from each category. Sample types 1–4 correspond to insulation layer damage, foreign body damage, insulator-conductor damage, and conductor-foreign body damage, respectively. The average results are reported in Table 2.

As shown in Table 2, training with the expanded dataset improves the detection performance for all four defect categories. The average precision increases from 83.45% to 94.33%, and the average recall increases from 80.25% to 92.93%, corresponding to improvements of 10.88 and 12.68 percentage points, respectively. Meanwhile, the average mAP50 increases from 83.45% to 94.65%, and the average mAP50–95 increases from 58.90% to 80.80%, with improvements of 11.20 and 21.90 percentage points, respectively. These results indicate that introducing synthetic images into the training set improves both defect recognition and localization performance, particularly under stricter IoU thresholds. Therefore, synthetic data augmentation can effectively alleviate the limitation caused by the relatively small number of real training samples in this study.

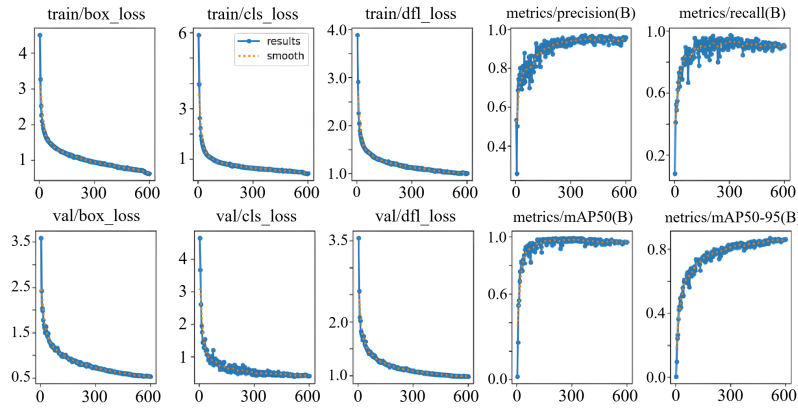
Table 2 Comparison of detection performance before and after dataset expansion.

Sample Type	Dataset	Precision(%)	Recall(%)	mAP50(%)	mAP50-95(%)
1	Before expansion	85.2	84.7	86.1	61.8
	After expansion	96.6	96.1	97.0	83.9
2	Before expansion	88.9	80.8	87.3	63.5
	After expansion	97.9	95.3	97.6	85.1
3	Before expansion	80.4	80.3	81.5	56.7
	After expansion	91.6	90.5	92.3	77.8
4	Before expansion	79.3	75.2	78.9	53.6
	After expansion	91.2	89.8	91.7	76.4

4.3 Analysis of Defect Detection Results

The proposed EMA-YOLOv8 model is trained and validated using the high-voltage cable defect dataset constructed in this study. To analyze the training process and convergence behavior of the model, the evolution of the box loss, classification-related loss, distribution focal loss, precision, and recall is examined, as shown in Fig. 11. The horizontal axis represents the number of training iterations, while the vertical axis represents the corresponding metric value.

As the training progresses, the loss values gradually decrease and tend to stabilize, while the precision and recall progressively improve and eventually reach relatively stable levels. These trends indicate that the optimization process converges steadily during training. The final detection performance of the model is further evaluated using the quantitative results on the real test set.

**Fig. 11** EMA-YOLOv8 model evaluation index results.

4.4 Ablation Experiment Analysis

Through a series of ablation experiments, the effectiveness of EfficientNet, the channel attention mechanism, and the box refinement strategy is evaluated for high-voltage cable defect detection. These evaluations are conducted within the framework of

the EMA-YOLOv8 network. The comparative experimental results are presented in Table 3, where CAM represents the channel attention mechanism, mAP represents the mean average precision, BR represents the box refinement strategy, PQ represents the parameter quantity, and PEI represents the precision efficiency index, which is defined as the ratio of mAP to PQ and is used to evaluate parameter utilization efficiency [39].

Table 3 Comparison of ablation experiment results.

Network	Group	EfficientNet	CAM	BR	mAP(%)	PQ(M)	PEI
YOLOv8	A				86.5	64.3	0.0135
	B	✓			88.0	44.7	0.0197
	C		✓		89.5	45.3	0.0198
	D	✓	✓	✓	96.1	46.1	0.0208

Through comprehensive analysis of the experimental results for Groups A–D, the following observations can be made. Group A employs the original YOLOv8 network and has the largest number of model parameters, with an mAP of 86.5%. Group B replaces the original CSPNet backbone with EfficientNet, increasing the mAP from 86.5% to 88.0% while reducing the number of model parameters from 64.3 M to 44.7 M. Group C further introduces the SE module into the backbone network, increasing the mAP by 1.5 percentage points to 89.5%, with the number of model parameters increasing by 0.6 M. By integrating EfficientNet, the channel attention mechanism, and the box refinement strategy, Group D achieves an mAP of 96.1%, corresponding to an increase of 6.6 percentage points compared with Group C. Meanwhile, the number of model parameters increases by 0.8 M to 46.1 M. In addition, the PEI values of Groups A–D are 0.0135, 0.0197, 0.0198, and 0.0208, respectively. Compared with Group A, the PEI of Group D increases from 0.0135 to 0.0208, indicating higher detection performance per unit of model parameters.

It is evident that, in comparison with the original YOLOv8 model, the EMA-YOLOv8 model proposed in this study demonstrates a remarkable enhancement in defect detection accuracy, a considerable reduction in model parameters, and improved parameter utilization efficiency.

4.5 Comparative Test Results and Analysis

To comprehensively evaluate the proposed method, EMA-YOLOv8 is compared with representative object detection models, including SSD, Faster-RCNN, YOLOv7, YOLOv8, YOLOv10, and RT-DETR-R34. For a fair comparison, all models are trained and evaluated using the same data partitions, input resolution, training budget (batch size, epoch, etc.), and hardware environment. Considering the differences in network architecture and training strategy, architecture-appropriate hyperparameter ranges are adopted instead of enforcing identical absolute values for all models. Specifically, the initial learning rate is searched over 5×10^{-4} , 1×10^{-3} , and 2×10^{-3} for SSD, Faster-RCNN, YOLOv8, and EMA-YOLOv8, and over 5×10^{-3} , 1×10^{-2} , and 2×10^{-2} for YOLOv7 and YOLOv10. Weight decay is evaluated at 1×10^{-4} and 5×10^{-4} . For

RT-DETR-R34, learning-rate scales of $0.5\times$, $1\times$, and $2\times$ the recommended reference setting are examined. All hyperparameters are selected exclusively on the validation set according to mAP50, and the real test set is used only for final evaluation. In particular, YOLOv8 and EMA-YOLOv8 use the same hyperparameter-search range and model-selection criterion to ensure a fair comparison. The comparative results are presented in Table 4.

Table 4 Comparison of results of different detection methods.

Model	Precision(%)	Recall(%)	F_1 (%)	GFLOPs(G)	Parameters(M)	mAP50(%)	FPS	Model Size(MB)
SSD	85.4	80.7	83.0	8.8	91.4	80.2	38	365.6
Faster-RCNN	88.1	84.2	86.1	14.5	103.2	83.5	25	412.8
YOLOv7	91.3	87.0	89.1	7.9	66.7	87.3	65	266.8
YOLOv8	92.0	87.5	89.7	7.6	64.3	88.1	72	257.2
YOLOv10	94.3	89.4	91.8	8.6	52.8	90.2	78	211.2
RT-DETR-R34	93.8	90.0	91.8	12.4	49.6	90.3	61	198.4
EMA-YOLOv8	95.5	89.2	91.3	7.2	46.1	90.5	85	184.4

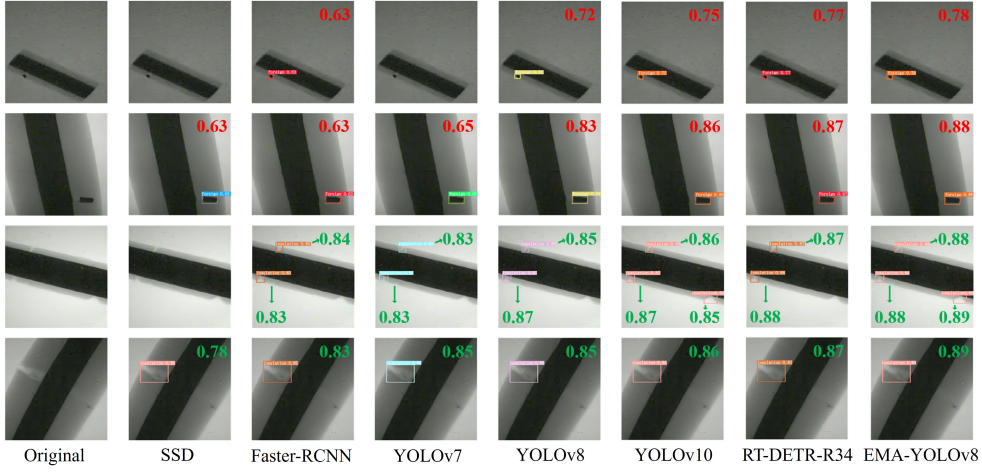


Fig. 12 Comparison of results of various defect detection methods.

As shown in Table 4, EMA-YOLOv8 achieves the highest precision and mAP50 among the compared methods, reaching 95.5% and 90.5%, respectively. Compared with the original YOLOv8, the precision and mAP50 are improved by 3.5 and 2.4 percentage points. Compared with YOLOv10 and RT-DETR-R34, EMA-YOLOv8 improves the precision by 1.2 and 1.7 percentage points and the mAP50 by 0.3 and 0.2 percentage points, respectively. In contrast, RT-DETR-R34 achieves the highest recall of 90.0%, while YOLOv10 and RT-DETR-R34 achieve the highest F_1 score of 91.8%. These results indicate that different models exhibit different precision-recall trade-offs, while EMA-YOLOv8 maintains competitive overall detection performance.

In terms of computational efficiency, EMA-YOLOv8 requires 7.2 GFLOPs and 46.1 M parameters, with a model size of 184.4 MB and an inference speed of 85 FPS. Compared with YOLOv8, the number of parameters and model size are reduced by 28.3%, while the inference speed is increased by 18.1%. Compared with YOLOv10, EMA-YOLOv8 reduces GFLOPs and parameters while increasing the inference speed from 78 to 85 FPS. Compared with RT-DETR-R34, the computational cost is also lower and the inference speed is significantly improved.

In addition to the quantitative comparison in Table 4, Fig. 12 provides qualitative detection results of seven representative methods, including SSD, Faster-RCNN, YOLOv7, YOLOv8, YOLOv10, RT-DETR-R34, and EMA-YOLOv8. It can be observed that SSD and Faster-RCNN tend to produce relatively lower-confidence predictions and are less stable in some defect cases. YOLOv7 and YOLOv8 improve the detection confidence and localization quality, but several defect instances are still detected with relatively moderate confidence scores. Compared with these methods, YOLOv10 and RT-DETR-R34 show more competitive qualitative results, with clearer localization and higher confidence in most cases. Among all compared methods, EMA-YOLOv8 yields relatively accurate localization and relatively high confidence scores for the displayed defect samples, indicating better visual detection performance on both foreign-body defects and insulation defects.

Overall, although YOLOv10 and RT-DETR-R34 show advantages in some individual metrics, the proposed EMA-YOLOv8 provides a favorable balance between detection accuracy, computational complexity, model size, and inference speed. In particular, its higher precision and mAP50, together with relatively low computational cost and high inference speed, make it suitable for real-time high-voltage cable X-ray defect detection.

4.6 Robustness Analysis Under X-Ray Image-Quality Perturbations

To evaluate the robustness of the proposed model to variations in X-ray image quality, controlled perturbations are applied to the real test set. Four representative types of image-quality degradation are considered, including brightness variation, contrast variation, Gaussian noise, and Gaussian blur. Representative examples are shown in Fig. 13, and the corresponding changes in mAP50 and mAP50–95 relative to the clean test set are reported in Table 5.

As shown in Table 5, the mAP50 varies within -1.24 to $+0.15$ percentage points under all tested perturbations, while the mAP50–95 shows a larger variation of -4.76 to $+1.16$ percentage points. The results indicate that the proposed model maintains relatively stable detection performance under the investigated X-ray image-quality perturbations, although relatively strong brightness and contrast changes have a greater influence on localization accuracy under stricter IoU thresholds.

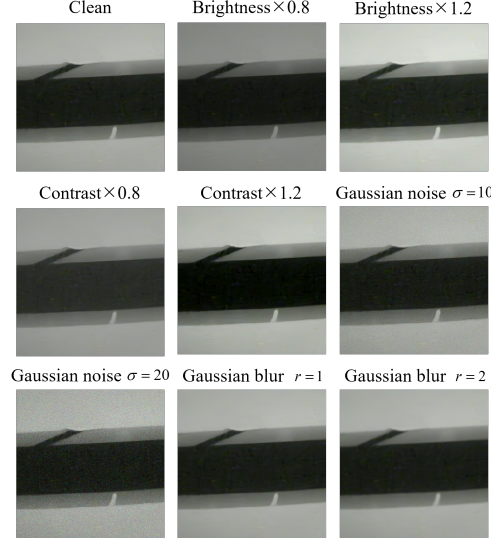


Fig. 13 Examples of X-ray images under different image-quality perturbations.

Table 5 Robustness evaluation under different X-ray image-quality perturbations.

Condition	Perturbation level	ΔmAP_{50} (%)	ΔmAP_{50-95} (%)
Clean	–	0.00	0.00
Brightness	0.8	-0.35	+1.16
	1.2	-0.34	-4.76
Contrast	0.8	-1.24	-4.29
	1.2	+0.15	-2.08
Gaussian noise	$\sigma = 10$	-0.13	+0.22
	$\sigma = 20$	-0.10	-0.74
Gaussian blur	$r = 1$	0.00	-0.11
	$r = 2$	-0.73	-2.22

5 Conclusion

In this paper, aiming at the problems of low accuracy, slow speed, and invasiveness in the internal defect detection of high-voltage cables with different sizes, the EMA-YOLOv8 model is proposed, and an X-ray-based NDT system is designed. By replacing CSPNet with EfficientNet as the backbone network and introducing the SE module, the feature extraction capability of the model is improved while the computational cost is reduced. Compared with the compared detection models, the number of operations is reduced by up to 7.3 GFLOPs, corresponding to a reduction of 50.3%, and the number of parameters is reduced by up to 57.1 M, corresponding to a reduction of 55.3%. At the same time, a box refinement strategy combining K-means initialization with IoU-guided matching is integrated with the multi-scale feature fusion mechanism based on the hybrid pyramid structure, enabling multi-directional transfer of feature and localization information at different scales and improving feature representation

and localization capability. Compared with the corresponding average values of the other detection models, EMA-YOLOv8 achieves improvements of 4.68, 2.73, 2.72, and 3.90 percentage points in precision, recall, F_1 score, and mAP50, respectively.

It is worth noting that the device designed in this paper has been successfully applied in the daily maintenance of the power grid in China, which significantly reduces the maintenance cost of the power grid and improves the detection efficiency. In the future, the research team will continue to focus on cable internal defect detection and explore more high-precision and low-cost detection methods.

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Data Availability. The dataset used in this study involves sensitive information related to the urban power grid system and therefore cannot be publicly shared at present.

Declarations

Competing Interests. The authors declare no competing interests.

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